

KUHMO KALLIOJOKI, Finland

WMO: 027990

Lat: 64.298N

Lon: 30.166E

Elev: 198

StdP: 98.96

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.4	-28.2	-34.8	0.1	-31.5	-31.2	0.2	-28.1	6.3	-2.4	5.8	-3.0	0.4	230	0.633

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.6	26.6	17.6	24.5	16.5	22.4	15.7	19.3	23.6	18.0	22.2	16.8	20.9	2.5	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.7	13.0	21.3	16.5	12.0	19.7	15.2	11.0	18.4	55.9	23.6	51.6	22.0	47.8	21.2	23.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5.8	5.2	4.5	-34.6	28.2	4.5	3.3	-37.8	30.5	-40.4	32.5	-42.9	34.3	-46.2	36.7
			-34.5	20.3	4.3	1.9	-37.6	21.7	-40.1	22.8	-42.5	23.9	-45.6	25.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	1.9	-12.1	-9.9	-6.1	0.5	7.7	12.1	15.8	13.6	8.8	2.0	-2.7	-7.3	(6)
(7)		DBStd	10.68	8.17	7.74	6.05	4.45	4.88	4.05	3.78	3.46	3.33	3.70	4.83	7.14	(7)
(8)		HDD10.0	3388	684	557	499	285	103	23	3	9	61	248	380	536	(8)
(9)		HDD18.3	6011	943	791	757	534	329	192	97	152	286	506	630	794	(9)
(10)		CDD10.0	447	0	0	0	1	32	86	183	120	26	1	0	0	(10)
(11)		CDD18.3	28	0	0	0	0	1	4	19	4	0	0	0	0	(11)
(12)		CDH23.3	305	0	0	0	0	24	61	174	46	0	0	0	0	(12)
(13)		CDH26.7	48	0	0	0	0	0	8	33	8	0	0	0	0	(13)
(14)	Wind	WSAvg	2.1	1.9	2.2	2.3	2.2	2.4	2.2	1.8	1.9	2.0	2.4	2.3	2.2	(14)
(15)	Precipitation	PrecAvg	609	39	31	30	32	51	60	74	79	62	52	49	47	(15)
(16)		PrecMax	815	76	58	55	68	87	98	146	138	101	121	92	109	(16)
(17)		PrecMin	440	11	9	12	7	14	25	38	31	16	5	19	8	(17)
(18)		PrecStd	101	18	15	11	16	24	21	31	31	25	30	18	26	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.9	3.6	7.3	16.6	25.9	27.9	29.3	27.6	20.1	11.7	7.3	2.7	(19)
(20)			MCWB	1.7	2.1	3.1	8.8	13.7	17.2	20.4	18.7	15.2	10.1	6.6	2.4	(20)
(21)		2%	DB	0.5	1.7	4.3	12.1	23.4	25.2	27.4	24.2	17.1	9.8	5.5	1.5	(21)
(22)			MCWB	0.3	1.1	1.2	6.6	13.2	16.6	18.6	17.1	12.8	8.5	5.2	1.2	(22)
(23)		5%	DB	-0.4	0.6	2.5	9.4	20.1	22.2	25.2	21.9	15.5	8.3	3.8	0.8	(23)
(24)			MCWB	-0.5	0.2	0.6	4.4	12.1	14.7	17.8	15.9	12.5	7.3	3.5	0.5	(24)
(25)		10%	DB	-1.5	-0.1	1.3	7.4	16.8	19.9	23.1	19.8	14.0	7.0	2.5	0.2	(25)
(26)			MCWB	-1.7	-0.4	0.1	3.2	10.1	13.4	16.8	15.3	11.8	6.3	2.3	0.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.6	2.4	3.4	9.6	15.4	19.5	21.6	20.0	15.9	10.7	7.1	2.6	(27)
(28)			MCDWB	1.9	3.0	6.1	14.2	22.1	23.5	27.9	25.7	19.0	11.4	7.2	2.6	(28)
(29)		2%	WB	0.2	1.2	1.9	7.1	14.1	17.7	20.0	17.9	14.2	8.7	5.2	1.2	(29)
(30)			MCDWB	0.3	1.7	3.5	11.9	20.2	23.2	24.3	21.6	16.1	9.2	5.5	1.3	(30)
(31)		5%	WB	-0.7	0.3	1.1	4.8	12.5	16.0	18.9	16.8	13.0	7.4	3.6	0.6	(31)
(32)			MCDWB	-0.5	0.6	2.1	8.0	19.4	20.5	23.1	20.7	14.6	8.0	3.8	0.7	(32)
(33)		10%	WB	-1.9	-0.5	0.4	3.8	10.7	14.5	17.7	15.9	12.1	6.3	2.3	0.0	(33)
(34)			MCDWB	-1.7	-0.2	1.2	6.1	16.0	18.3	21.7	19.2	13.6	7.0	2.4	0.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.0	7.8	10.0	9.8	10.5	9.9	9.6	9.0	7.1	4.7	4.0	5.3	(35)
(36)			MCDBR	4.0	4.8	8.3	13.4	16.0	14.1	12.7	12.8	9.5	6.4	4.1	3.7	(36)
(37)			MCWBR	4.1	4.6	6.4	7.9	7.3	6.4	5.3	6.1	5.8	5.3	4.1	3.5	(37)
(38)		5% WB	MCDWB	4.0	5.0	6.4	11.4	14.4	11.6	11.4	10.4	7.9	5.8	4.1	3.4	(38)
(39)			MCWBR	4.0	4.8	5.3	7.1	7.2	6.0	5.4	5.5	5.2	5.2	4.1	3.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.209	0.274	0.303	0.337	0.345	0.342	0.367	0.346	0.323	0.300	0.359	0.202	(40)	
(41)		taud	1.949	2.101	2.152	2.257	2.453	2.487	2.435	2.475	2.526	2.386	2.657	1.953	(41)	
(42)		Ebn at Noon	339	620	771	833	867	877	838	825	773	613	321	125	(42)	
(43)		Edh at Noon	27	62	92	105	95	94	97	85	66	48	24	11	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.15	0.70	2.17	3.76	4.62	5.14	4.83	3.60	2.01	0.72	0.16	0.05	(44)	
(45)		RadStd	0.02	0.14	0.22	0.36	0.54	0.34	0.52	0.35	0.22	0.11	0.03	0.01	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (22 neighbors)	+0.54	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page